ASR-12 Primary Surveillance Radar

Representative parameter datasheet for MATLAB agentic-workflow exercises

Revision: 1.2 - 8 October 2026
Classification: Training Use Only

IMPORTANT - SIMULATION DATA ONLY.

This is a fictional, representative training artifact for classroom radar modeling. Values, site name, and configuration are not manufacturer, FAA, airport, or operational specifications and must not be used for navigation, procurement, safety, or real-world system configuration.

Terminal-area primary surveillance

Rotating S-band radar configuration designed as a coherent parameter set for a classroom air-traffic-control scenario and a first-pass MATLAB radar simulation.

240	60	2.8
NMI INSTRUMENTED RANGE	RPM SCAN RATE	GHZ NOMINAL FREQUENCY

Configuration at a glance

Radar role	**Primary surveillance**
Scenario ID	**KMTN-TRN-01**
Coverage mode	**Terminal / 360 deg**
Propagation model	**Free-space baseline**
Target class	**Fixed-wing aircraft**
Configuration	**Baseline terminal coverage**

1. System identity and operating envelope

Parameter	Representative value
System designation	ASR-12 (training representative)
Radar type	Coherent pulsed primary surveillance radar
Frequency band	S-band
Instrumented range	0-240 nmi (0-444 km)
Azimuth coverage	0-360 deg
Elevation coverage	20 deg
Environmental baseline	Standard atmosphere; clear air

2. Primary model input set

Parameter	Representative value
Carrier frequency, fc	2.80 GHz
Peak transmit power, Pt	1.10 MW
Pulse repetition frequency, PRF	325 Hz
Transmit pulse width, tau	2.15 us
Antenna peak gain, G	37.21 dBi
Azimuth 3 dB beamwidth	1.4 deg
Elevation 3 dB beamwidth	4.5 deg
Antenna scan rate	60 rpm
Receiver noise figure	4.0 dB
System losses	6.0 dB
Reference target RCS	10 m2

ASR-12 Modeling Notes

3. Detection and measurement assumptions

Parameter	Representative value	Use in exercise
Probability of detection, Pd	0.90 at reference condition	Reference only: 10 m2 target at 240 nmi, boresight, clear air.
Probability of false alarm, Pfa	1.0e-6 per resolution cell	Set detector threshold or use as a stated evaluation constraint.
Range measurement accuracy	+/-116 m (exercise assumption)	Add zero-mean Gaussian range error for track reports.
Azimuth measurement accuracy	+/-0.176 deg (exercise assumption)	Add zero-mean Gaussian bearing error for track reports.
Range resolution (nominal)	323 m	Use c*tau/2; no pulse compression in baseline configuration.
Unambiguous range (baseline)	461 km (249 nmi)	At 325 Hz; exceeds the 240 nmi reporting gate.
Update interval	1.0 s	One measurement opportunity per antenna revolution.

4. Classroom air-traffic-control scenario

Radar site

Fictional KMTN Training Terminal, reference position [0, 0, 0] m in local ENU coordinates. Antenna phase center height: 10 m above ground level.

Targets at scenario start

Track	Position [E, N, U] m	Velocity [E, N, U] m/s	RCS
TRN101	[18,000, 65,000, 2,800]	[-115, -40, 0]	10 m2
TRN205	[-32,000, 48,000, 4,100]	[95, -65, -2]	6 m2
TRN317	[7,000, 26,000, 1,700]	[45, 130, 3]	3 m2

Exercise scope

- Import this document and extract a structured parameter set.
- Build a rotating monostatic radar model.
- Simulate range/azimuth detections for the three targets.
- Compare ideal versus noisy measurement reports over 60 s.

Deliberately excluded from baseline

secondary surveillance · IFF · terrain masking · weather clutter · multipath · target fluctuation

5. Data-quality notes

Rule	Expected behavior
Operating units	Values are stated in the customary units used in this reference document.
Derived quantities	Range resolution, unambiguous range, and scan period are nominal values based on the listed baseline configuration.
Configuration control	Changes to the listed values require a corresponding revision to this training reference.
Document status	Fictional training reference; not an operational, procurement, or maintenance specification.

The 325 Hz baseline waveform gives approximately 461 km (249 nmi) unambiguous range, exceeding the 240 nmi instrumented-range requirement.